

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Full Stack development MERN Practical

Module - II

Practical –VIII

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Class:TYIT	Batch:2
Date of Assignment: 23/09/26	Date/Time of Submission: 30/09/26

8.API Creation and Integration.

a. Create a simple REST API using Express.

CODE:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
.then(() => {
console.log("MongoDB Connected Successfully");})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);});
// Simple REST API
app.get("/", (req, res) => {
res.send("Hey T054 Student Management API is Running");
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);});
```

OUTPUT:

```
PS C:\Users\mvlu\Desktop\MERN PRAC\MongoDBPractical> npm install express mongoose cors
added 69 packages, and audited 89 packages in 7s

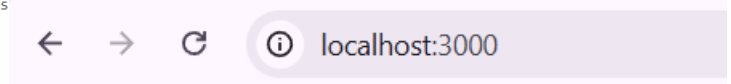
30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\mvlu\Desktop\MERN PRAC\MongoDBPractical> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```

b. Implement a GET API to retrieve data.

CODE:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
.then(() => {
console.log("MongoDB Connected Successfully");})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);});
```



Hey T054 Student Management API is Running

```
// Simple REST API
app.get("/", (req, res) => {
  res.send("Hey T054 Student Management API is Running");
});
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("Students")
      .find()
      .toArray();
    res.json(students);
  } catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

c. Fetch data from the API in a React application.

CODE:

Service.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
app.use(cors());
app.use(express.json());
// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => { console.log("MongoDB Connected Successfully"); })
  .catch((error) => { console.log("MongoDB Connection Failed"); });
// Home
app.get("/", (req, res) => {
  res.send(" Hey T054 Student Management API is Running");
});
// Get Students
app.get("/students", async (req, res) => {
  try {const students = await mongoose.connection.db
    .collection("students")
    .find({})
    .toArray();
    res.json(students); }
  catch (error) {res.status(500).json({
    message: "Error fetching students"
  });}
});
// Add Student
app.post("/students", async (req, res) => {
  try {const student = req.body;
    await mongoose.connection.db
      .collection("Students")
      .insertOne(student);
    res.json({
      message: "Student added successfully"
    });
  } catch (error) {res.status(500).json({
    message: "Error adding student"
  });}
});
// Start Server
app.listen(PORT, () => {console.log(`Server is running on http://localhost:${PORT}`);});
```

App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => {
    axios.get("http://localhost:3000/students")
```

OUTPUT:

```
[
  {
    "_id": "6ab3cee70af4ad043b3ce646",
    "rollNo": 55,
    "name": "Rahul Sharma",
    "course": "Information Technology",
    "marks": 85
  },
  {
    "_id": "6ab3cee70af4ad043b3ce647",
    "rollNo": 56,
    "name": "Priya Patil",
    "course": "Information Technology",
    "marks": 92
  },
  {
    "_id": "6ab3cee70af4ad043b3ce648",
    "rollNo": 57,
    "name": "Amit Joshi",
    "course": "B.Sc. CS",
    "marks": 78
  },
  {
    "_id": "6ab3dde0af4ad043b3ce649",
    "rollNo": 54,
    "name": "Priti Yadav",
    "course": "B.Sc. IT",
    "marks": 78
  }
]
```

```

.then((response) => {
  setStudents(response.data);    })
.catch((error) => {
  console.log(error); }); }, []);
return (
  <div>
    <h1>Student List</h1>
    <table border="1">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>
      <tbody>
        {students.map((student) => (
          <tr key={student._id}>
            <td>{student.rollNo}</td>
            <td>{student.name}</td>
            <td>{student.course}</td>
            <td>{student.marks}</td>
          </tr>))}
      </tbody>
    </table>
  </div>);}
export default App;

```

OUTPUT:

```

PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac> cd my-react
PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm install axios

up to date, audited 169 packages in 5s

37 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm run dev

> my-react@0.0.0 dev
> vite

VITE v8.3.0 ready in 541 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help

PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> cd..
PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac> node service.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
PS C:\Users\priti_yadav\OneDrive\Desktop\Mern Practical\Exprac> node service.js
Server is running on http://localhost:3000
MongoDB Connected Successfully

```

← localhost:5173

Student List

Roll No	Name	Course	Marks
55	Rahul Sharma	Information Technology	85
56	Priya Patil	Information Technology	92
57	Amit Joshi	B.Sc. CS	78
54	Priti Yadav	B.Sc. IT	78

d. Send data from React to the server using a POST request.

CODE:

App.jsx:

```

import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get Students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);    })
      .catch((error) => {
        console.log(error);    });    });
    useEffect(() => {

```

```

    getStudents();  }, []);
// Add Student
const addStudent = () => {
    const newStudent = {
        rollNo: Number(rollNo),
        name: name,
        course: course,
        marks: Number(marks)    };
    axios.post("http://localhost:3000/students", newStudent)
        .then((response) => {
            alert(response.data.message);
            setRollNo("");
            setName("");
            setCourse("");
            setMarks("");

            getStudents();    })
        .catch((error) => {
            console.log(error);    });    };
return (
    <div>
        <h1>Student Management System</h1>
        <h2>Add Student</h2>
        <input
            type="number"
            placeholder="Roll No"
            value={rollNo}
            onChange={(e) => setRollNo(e.target.value)}    />
        <br /><br />
        <input
            type="text"
            placeholder="Student Name"
            value={name}
            onChange={(e) => setName(e.target.value)}    />
        <br /><br />
        <input
            type="text"
            placeholder="Course"
            value={course}
            onChange={(e) => setCourse(e.target.value)}        />
        <br /><br />
        <input
            type="number"
            placeholder="Marks"
            value={marks}
            onChange={(e) => setMarks(e.target.value)}        />
        <br /><br />
        <button onClick={addStudent}>
            Add Student
        </button>
        <hr />
        <h2>Student List</h2>
        <table border="1" cellPadding="10">
            <thead>
                <tr>
                    <th>Roll No</th>
                    <th>Name</th>
                    <th>Course</th>
                    <th>Marks</th>
                </tr>
            </thead>
            <tbody>

```

```

        {students.map((student) => (
            <tr key={student._id}>
                <td>{student.rollNo}</td>
                <td>{student.name}</td>
                <td>{student.course}</td>
                <td>{student.marks}</td>
            </tr>
        ))}
    </tbody>
</table>
</div> );}

export default App;
Service.js:
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
app.use(cors());
app.use(express.json());
// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
    .then(() => {
        console.log("MongoDB Connected Successfully");
    })
    .catch((error) => {
        console.log("MongoDB Connection Failed");
        console.log(error);
    });
// Home Route
app.get("/", (req, res) => {
    res.send("Student Management API is Running");
});
// GET - Display Students
app.get("/students", async (req, res) => {
    try {
        const students = await mongoose.connection.db
            .collection("students")
            .find()
            .toArray();
        res.json(students);
    } catch (error) {
        res.status(500).json({
            message: "Error fetching students"
        });
    }
});
// POST - Add Student
app.post("/students", async (req, res) => {
    try {
        const student = req.body;
        await mongoose.connection.db
            .collection("students")
            .insertOne(student);
        res.json({
            message: "Student added successfully",
            student: student
        });
    } catch (error) {
        res.status(500).json({
            message: "Error adding student"
        });
    }
});
// Start Server
app.listen(PORT, () => {
    console.log(`Server is running on http://localhost:${PORT}`);
});

```

OUTPUT:

```

PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm run dev
> my-react@0.0.0 dev
> vite

VITE v8.3.0 ready in 566 ms
  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
o

```

```

PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac> node service.js
Server is running on http://localhost:3000
MongoDB Connected Successfully

```

Student Management System

Add Student

Add Student

Student List

Roll No	Name	Course	Marks
55	Rahul Sharma	Information Technology	85
56	Priya Patil	Information Technology	92
57	Amit Joshi	B.Sc. CS	78
54	Priti Yadav	B.Sc. IT	78

Stude

localhost:5173 says
Student added successfully

OK

Add Student

Student Management System

Add Student

Add Student

Student List

Roll No	Name	Course	Marks
55	Rahul Sharma	Information Technology	85
56	Priya Patil	Information Technology	92
57	Amit Joshi	B.Sc. CS	78
54	Priti Yadav	B.Sc. IT	78
19	Prashant Jha	B.COM	95

e. Store data in MongoDB.

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios
      .get("http://localhost:3000/students")
      .then((response) => {
        console.log("Student Data:", response.data);
        setStudents(response.data);
      })
      .catch((error) => {
        console.log("Fetch Error:", error);
      });
  }; // Fetch data when page loads
  useEffect(() => {
    getStudents();
  }, []);
  // Add student
  const addStudent = () => {
    if (!rollNo || !name || !course || !marks) {
      alert("Please fill all fields");
      return;
    }
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios
      .post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
      });
  };
  // Clear form
  setRollNo("");
  setName("");
  setCourse("");
  setMarks("");
  // Refresh student list
  getStudents();
}
.catch((error) => {
  console.log("POST Error:", error);
  alert("Error adding student");
});
return (
  <div>
    <h1>Student Management System</h1>
    <h2>Add Student</h2>
    <input
      type="number"
      placeholder="Roll No"
      value={rollNo}
      onChange={(e) => setRollNo(e.target.value)} />
    <br />
    <input
      type="text"
      placeholder="Student Name"
      value={name}
      onChange={(e) => setName(e.target.value)} />
    <br />
    <input
      type="text"
      placeholder="Course"
      value={course}
      onChange={(e) => setCourse(e.target.value)} />
    <br />
    <input
      type="number"
      placeholder="Marks"
      value={marks}
      onChange={(e) => setMarks(e.target.value)} />
    <br />
    <button onClick={addStudent}>
      Add Student
    </button>
    <hr />
    <h2>Student List</h2>
    <table border="1" cellPadding="10">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>
      <tbody>
        {students.map((student) => (
          <tr key={student._id}>
            <td>{student.rollNo}</td>
            <td>{student.name}</td>
            <td>{student.course}</td>
            <td>{student.marks}</td>
          </tr>
        ))}
      </tbody>
    </table>
  </div>
);
export default App;
```

```
<br />
<input
  type="text"
  placeholder="Course"
  value={course}
  onChange={(e) => setCourse(e.target.value)} />
<br />
<input
  type="number"
  placeholder="Marks"
  value={marks}
  onChange={(e) => setMarks(e.target.value)} />
<br />
<button onClick={addStudent}>
  Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1" cellPadding="10">
  <thead>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
    </tr>
  </thead>
  <tbody>
    {students.map((student) => (
      <tr key={student._id}>
        <td>{student.rollNo}</td>
        <td>{student.name}</td>
        <td>{student.course}</td>
        <td>{student.marks}</td>
      </tr>
    ))}
  </tbody>
</table>
</div> ); }
export default App;

server.js :
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.error(error);
  });
// Home API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
// GET API - Fetch students
app.get("/students", async (req, res) => {
```

```

try { const students = await mongoose.connection.db
.collection("Students")
.find()
.toArray();
res.json(students);
} catch (error) {
console.error(error);
res.status(500).json({
message: "Error fetching students",
error: error.message }); } });
// POST API- Store students
app.post("/students", async (req, res) => {
try {
const student = req.body;
await mongoose.connection.db

```

OUTPUT:-

```

.collection("Students")
.insertOne(student);
res.json({
message: "Student added successfully",
student: student });
} catch (error) {
console.error(error);
res.status(500).json({
message: "Error adding student",
error: error.message }); } });
// Start Server
app.listen(PORT, () => {
console.log(`Server is running on
http://localhost:${PORT}`); });

```

Roll No	Name	Course	Marks
55	Rahul Sharma	Information Technology	85
56	Priya Patil	Information Technology	92
57	Amit Joshi	B.Sc. CS	78
54	Priti Yadav	B.Sc. IT	78

f. Display stored data in React page.

OUTPUT:

Student List

Roll No	Name	Course	Marks
55	Rahul Sharma	Information Technology	85
56	Priya Patil	Information Technology	92
57	Amit Joshi	B.Sc. CS	78
54	Priti Yadav	B.Sc. IT	78
19	Prashant Jha	B.COM	95